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Where Data Lives: Disks and File Systems



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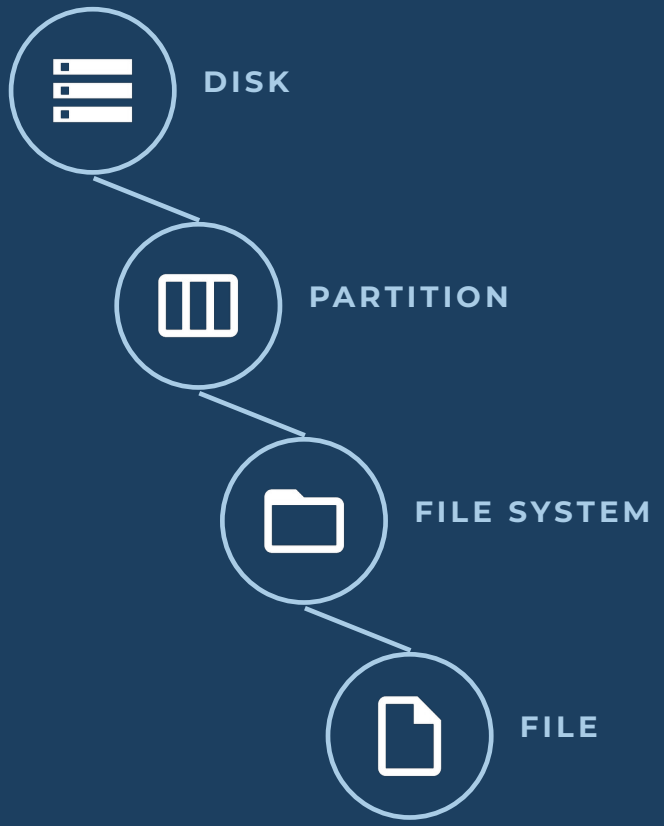
UTM BIG DATA CENTRE
TOGETHER WE MAKE MOTIONS...

Menginovasi Penyelesaian Lestari

DFBOOK DIGITAL FORENSICS · LESSON 4.0

Where Data Lives: Disks and File Systems

Lesson overview: why deleted and hidden data can still be found



LESSON 4.0 · OVERVIEW

Disks and File Systems

Why deleted and hidden data can still be found

BY THE END YOU WILL BE ABLE TO

Describe sectors, clusters and partitions

And how a file occupies them

Explain how FAT and NTFS track files

Names, locations and timestamps, at concept level

Explain slack space and deleted files

And recover a deleted file in the lab

Find key Windows artefacts

Recycle Bin, browser history, Registry and memory files

LESSON 4.0 · ROUTE

Four Sub-Lessons

Two demos and two hands-on labs

4.1	**Sectors, Clusters and Partitions**	Demo: partitions in Autopsy
4.2	**How FAT and NTFS Keep Track of Files**	Demo: one file, two file systems
4.3	**Slack Space and Deleted Files**	Lab: recover a deleted invoice
4.4	**Windows Artefacts Investigators Use**	Lab: Recycle Bin and browser history

Labs use the training image only. Work on the verified copy, never on an original or on real evidence.

LESSON 4.0 · THE CASE

The Teratai Case So Far

From a verified image to what is inside it

SO FAR

The laptop and USB drive were imaged through a write-blocker, and the hashes matched.

NOW

The examiner opens the verified image to find where the evidence is stored, and what was deleted.

QUESTIONS FOR THE IMAGE

- **How is the drive divided up?**
- **How are files recorded?**
- **Were invoices deleted?**
- **What did the user do?**

The question for this lesson: where on a drive does data live, and why does deleting it rarely remove it?

UP NEXT

Sectors, Clusters and Partitions

We start with how a drive is laid out, from the smallest unit to the partition.

- **Inside a disk drive**
- **Clusters and addresses**
- **Partitions**

DFBOOK DIGITAL FORENSICS · LESSON 4.1

Sectors, Clusters and Partitions

How a drive is laid out, and where a file sits

IN THIS LESSON

1. Inside a disk drive
2. Clusters and addresses
3. Partitions



TRACKS AND SECTORS

LESSON 4.1 · OVERVIEW

Sectors, Clusters and Partitions

To find evidence, you need to know where a drive stores data. This lesson works up from the smallest unit, the sector, to the partition.

BY THE END YOU WILL BE ABLE TO

1. **Describe the parts of a disk drive**
2. **Explain clusters and how a file occupies them**
3. **Describe partitions, and where data can hide**

INTRODUCTION · WHY THIS MATTERS

Why This Matters

Illustrative: a 2 TB drive that the computer reports as 1.5 TB

1.5 TB: what the operating system sees	500 GB unseen

THE WHOLE DRIVE: 2 TB

Hidden at the end

The drive is set up to declare less space than it has.

Hidden between partitions

A partition removed from the partition table, but not erased.

Forensic tools can find these areas in a whole-drive image, if you know to look.

INTRODUCTION · ROADMAP

Your Route Through Lesson 4.1

Each part answers one question, in order

1

PART 1

Inside a disk drive

What are tracks, cylinders and sectors?

2

PART 2

Clusters and addresses

How does a file occupy space on a drive?

3

PART 3

Partitions

How is a drive divided, and where can data hide?

One case runs through the module: Teratai Supplies, where auditors have found payments for materials that were never delivered.

PART 1 OF 3

Inside a Disk Drive

The physical layout behind every file

01

IN THIS PART YOU WILL LEARN TO

- Name the parts of a disk drive
- Calculate a drive's size from its geometry

KEY TERMS

Track **Sector** **Cylinder**

PART 1 · STRUCTURE

Parts of a Disk Drive

Hard disk drives store data on spinning platters



SSDs have no platters, but present the same layout to the computer.

Track	One of the rings on a platter where data is stored.
Sector	A section of a track: usually 512 to 4,096 bytes.
Cylinder	The same track on every platter, stacked as a column.
Head	Reads and writes data; two per platter, top and bottom.
Geometry	The layout of platters, tracks and sectors.

PART 1 · WORKED EXAMPLE

Working Out a Drive's Size

Illustrative: an older drive described by its geometry

THE GEOMETRY

Cylinders	1,024
Heads	32
Sectors per track	63
Bytes per sector	512

1 Count the sectors

`1,024 x 32 x 63 = 2,064,384 sectors`

2 Multiply by sector size

`2,064,384 x 512 = 1,056,964,608 bytes`

3 Express it simply

`About 1.06 GB`

ANSWER

Cylinders x heads x sectors x bytes per sector: about 1.06 GB. This is the CHS calculation.

PART 2 OF 3

Clusters and Addresses

How the operating system groups sectors to store files

02

IN THIS PART YOU WILL LEARN TO

- Explain what a cluster is
- Describe how a file occupies clusters

KEY TERMS

Cluster **Logical address** **Fragmentation**

PART 2 · CONCEPT

Sectors and Clusters

The drive stores sectors; the operating system works in clusters

Cluster: a storage unit of one or more sectors, set when a partition is formatted. Grouping sectors reduces the work of reading and writing.

Logical address

The cluster's number within a partition. FAT counts from 2, NTFS from 0.

Physical address

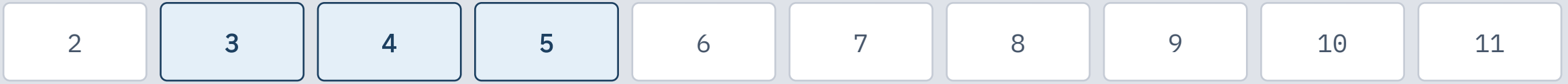
The sector's number on the whole drive, from sector 0 to the last.

PART 2 · SEQUENCE

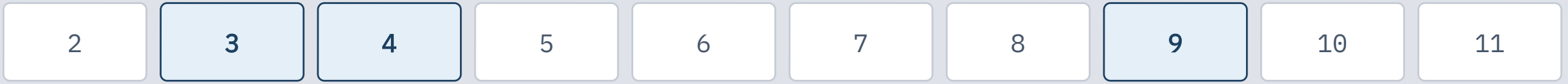
How a File Occupies Clusters

Whole clusters are allocated, and chained together as the file grows

A FILE IN ONE PIECE



THE SAME FILE, FRAGMENTED



Cluster numbers shown in each box; shaded clusters belong to the file.

Fragmentation: a file stored in clusters that are not next to each other. In FAT, each entry points only forward, to the next cluster.

PART 2 · WORKED EXAMPLE

How Many Clusters for One Invoice?

Illustrative: a 10,000-byte invoice on a drive with 4 KB (4,096-byte) clusters

1 Divide by the cluster size

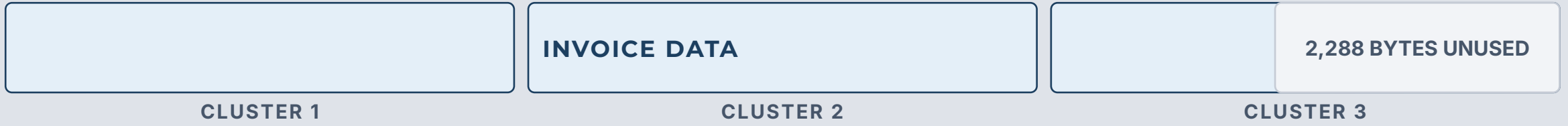
`10,000 / 4,096 = 2.44`

2 Round up to whole clusters

`3 clusters = 12,288 bytes`

3 Find the unused space

`12,288 - 10,000 = 2,288 bytes`



ANSWER

Three clusters, with 2,288 bytes left over. That leftover space is called slack, covered in Lesson 4.3.

PART 3 OF 3

Partitions

How a drive is divided, and where data can hide

03

IN THIS PART YOU WILL LEARN TO

- Compare MBR and GPT drives
- Spot hidden or deleted partitions

KEY TERMS

Partition **MBR** **GPT**

PART 3 · COMPARISON

MBR and GPT

Two ways to record a drive's partitions, both starting at sector 0

	Master Boot Record (MBR)	GUID Partition Table (GPT)
Maximum drive size	About 2.2 TB	About 9.4 ZB (zettabytes)
Sector addressing	Cylinder, head and sector	Logical block address (LBA), from 0
Partition entries	Up to three primary, plus one extended	Many more
Needs	Any PC	Firmware with EFI support

Partition: a section of a drive that holds a logical drive, such as C: in Windows. One drive can hold several.

PART 3 · STRUCTURE

Reading an MBR Partition Table

Four entries, each with a code that names the file system

ENTRY START (IN SECTOR 0)	
0x1BE	1st primary partition
0x1CE	2nd primary partition
0x1DE	3rd primary partition
0x1EE	Extended partition

SOME TYPE CODES (HEXADECIMAL)	
07	NTFS or exFAT
0B, 0C	FAT32
05, 0F	Extended partition
17, 1B	Hidden NTFS, hidden FAT32

Each entry also records where its partition starts, and how many sectors it uses. Hexadecimal values are written with 0x in front.

PART 3 · CATEGORIES

Hiding Data in Partitions

Three tricks, and why imaging the whole drive defeats them

The partition gap

Small unused space between partitions, made larger to hold data.

A hidden partition

Its entry removed from the partition table, so the system cannot see it.

A shrunken drive

Declared smaller than it is, leaving space at the end.

Most forensic tools detect hidden partitions in an image of the whole drive. A disk editor such as HxD can read them too.

PART 3 · WORKED EXAMPLE

Three Partition Entries

Illustrative: type codes read from a drive's partition table

Entry	Type code	Meaning
1st	07	NTFS
2nd	0B	FAT32
3rd	**17**	**Hidden NTFS**

Which partition needs a closer look, and why?

Code 17 marks an NTFS partition set to be hidden from the user.

ANSWER

The third: someone chose to hide it. Examine its contents, and record where it starts and ends.

PART 3 · DEMO

Partitions in Autopsy

Demo: the partitions of the Teratai training image

Add the image

Autopsy reads the verified E01 image.

List the volumes

Each partition appears as a volume, with its start and end sectors.

Check the gaps

Unallocated volumes may hold hidden or deleted partitions.

Volume: how Autopsy and other tools show a partition, or an unallocated area, in an image.

Account for all the space on the drive: any area you cannot explain needs a closer look.

LESSON 4.1 · SUMMARY

Lesson Summary

Three ideas that lead to one habit

1

Inside a disk drive

Platters, tracks, cylinders and sectors of 512 to 4,096 bytes; size by the CHS calculation.

2

Clusters and addresses

Sectors grouped into clusters; files take whole clusters, and may be fragmented.

3

Partitions

MBR or GPT tables in sector 0; type codes; gaps and hidden partitions.

Account for every sector of the drive.

UP NEXT

How FAT and NTFS Keep Track of Files

Next: how file systems record each file's name, location and timestamps.

- **File systems and FAT**
- **NTFS and the MFT**
- **Inside an MFT record**

DFBOOK DIGITAL FORENSICS · LESSON 4.2

How FAT and NTFS Keep Track of Files

Where a file's name, location and dates are recorded

IN THIS LESSON

1. File systems and FAT
2. NTFS and the MFT
3. Inside an MFT record



ONE FILE'S RECORD

LESSON 4.2 · OVERVIEW

How File Systems Track Files

A file system is the drive's index. It records where each file is, what it is called, and when it was created and changed. That index is where examiners look first.

BY THE END YOU WILL BE ABLE TO

1. Explain what a FAT entry records
2. Describe NTFS and its Master File Table
3. Explain where NTFS keeps names and timestamps

INTRODUCTION · WHY THIS MATTERS

Why This Matters

Three questions about every file, answered by the file system

What is it called?

The file's name, which may have changed.

Where is it stored?

The clusters that hold its contents.

When was it used?

Dates and times it was created, changed and opened.

These answers are not kept inside the file. They are kept in the file system's own records, which most users never see.

Know where those records live, and you can find, date and recover files.

INTRODUCTION · ROADMAP

Your Route Through Lesson 4.2

Each part answers one question, in order

1

PART 1

File systems and FAT

What does a FAT file system record?

2

PART 2

NTFS and the MFT

How does NTFS keep track of files?

3

PART 3

Inside an MFT record

Where are a file's name, contents and dates?

One case runs through the module: Teratai Supplies, where auditors have found payments for materials that were never delivered.

PART 1 OF 3

File Systems and FAT

The older, simpler file system still found on USB drives

01

IN THIS PART YOU WILL LEARN TO

- Explain what a FAT entry records
- Follow a file's chain of clusters

KEY TERMS

File system **FAT** **Attribute**

PART 1 · STRUCTURE

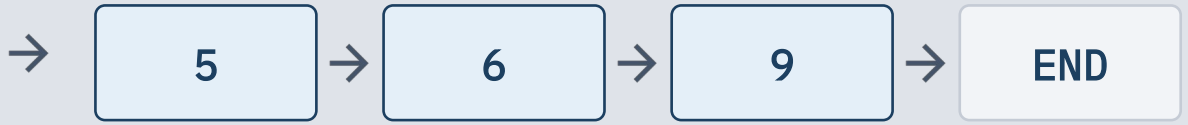
What FAT Records

A directory entry for each file, and a table that chains its clusters

DIRECTORY ENTRY

Name	File and folder names
Dates	Date and time stamps
Start	The first cluster of the file
Attributes	Archive, hidden, system, read-only

THE FILE ALLOCATION TABLE



Each cluster's entry points to the next one in the file.

File Allocation Table (FAT): the file system Microsoft first designed for floppy disks, still used on USB drives and memory cards.

PART 1 · COMPARISON

Versions of FAT

From floppy disks to memory cards

Version	Where it is used
FAT12	Floppy disks, and drives up to 16 MB
FAT16	Older Windows systems; partitions up to 4 GB
FAT32	Larger drives than FAT16 could handle
exFAT	Flash drives and memory cards; very large files such as video
VFAT	An extension that allows long file names

Early FAT allowed only 8-character names with 3-character extensions, such as INVOICE.PDF.

PART 1 · WORKED EXAMPLE

Following a FAT Chain

Illustrative: an invoice on the Teratai USB drive

WHAT IS RECORDED

Directory entry

`INV1042.PDF, starts at cluster 5`

FAT entry 5

`points to 6`

FAT entry 6

`points to 9`

FAT entry 9

`end of file`

1. **Which clusters hold it?**
   Clusters 5, 6 and 9.
2. **Is it in one piece?**
   No: it jumps from 6 to 9, so it is fragmented.
3. **What if the chain breaks?**
   FAT points only forward, so later clusters can be lost.

ANSWER

Clusters 5, 6 and 9, in that order. A forensic tool rebuilds the file by following the chain.

PART 2 OF 3

NTFS and the MFT

The main file system in modern Windows

02

IN THIS PART YOU WILL LEARN TO

- Describe what NTFS adds to FAT
- Name key NTFS system files

KEY TERMS

NTFS | MFT | Journaling

PART 2 · CONCEPT

What NTFS Adds

The NT File System, used from Windows NT to Windows 11

More about each file

Security settings, ownership and other attributes.

Journaling

Records each change before making it, so it can recover after a failure.

Smaller clusters

4 KB on most drives up to 16 TB, so less wasted (slack) space.

Unicode names

Supports any language; search for text in both encodings.

Master File Table (MFT): the first file on an NTFS partition, with a record for every file and folder. It starts at about 12.5% of the disk and can grow to 50%.

PART 2 · CATEGORIES

NTFS System Files

On NTFS, everything is a file, including the file system's own records

System file	What it holds
$MFT	A record for every file and folder on the partition
$LogFile	Recent transactions, for recovery after a failure
$Bitmap	A map of which clusters are in use and which are free
$I30	A folder's index, when its MFT record runs out of room
$UsnJrnl	A journal of changes made to files

These files are hidden in File Explorer. View them with forensic tools, such as Autopsy or FTK Imager.

PART 3 OF 3

Inside an MFT Record

Where NTFS keeps a file's name, contents and dates

03

IN THIS PART YOU WILL LEARN TO

- Describe the parts of an MFT record
- Compare FAT and NTFS records

KEY TERMS

Attribute **Resident** **Timestamp**

PART 3 · STRUCTURE

One MFT Record

1,024 bytes for each file, made of a header and attributes

FILE Header	0x10 Standard information	0x30 File name	0x80 Data

Header	Starts with the word FILE, and says where the first attribute begins.
Standard information (0x10)	The file's dates and times.
File name (0x30)	The file's name, with a second set of dates and times.
Data (0x80)	The contents of a small file, or a list of where a larger file is stored.

PART 3 · CONTRAST

Small Files and Large Files

Where the data attribute keeps a file's contents

RESIDENT

Inside the record

- A very small file's contents sit in the MFT record itself
- No separate clusters are used

NONRESIDENT

Out on the disk

- The record holds a list of the clusters used
- This list is called a run list

When a file is deleted, its record is marked free and its run list removed, but the contents often stay on the disk.

PART 3 · CONCEPT

Timestamps

The dates and times a file system records

Created

When the file was made on this volume

Modified

When its contents last changed

Accessed

When it was last opened (not always updated)

Record changed

When its NTFS record last changed (NTFS only)

FAT Keeps created and modified times, and a last-accessed date, in the directory entry.

NTFS Keeps two sets: in standard information (0x10) and in file name (0x30). If they disagree, find out why.

PART 3 · WORKED EXAMPLE

One Invoice, Two File Systems

Illustrative: where each detail of INV1042.PDF is recorded

Detail	On a FAT drive	On an NTFS drive
Name	Directory entry	File name attribute (0x30)
Location	Starting cluster, then the FAT chain	Data attribute (0x80): resident or run list
Timestamps	Directory entry	Attributes 0x10 and 0x30

ANSWER

The same three facts, kept in different places. NTFS keeps more, including two sets of timestamps.

LESSON 4.2 · SUMMARY

Lesson Summary

Three ideas that lead to one habit

1

File systems and FAT

Directory entries hold names, dates, start cluster and attributes; the table chains clusters.

2

NTFS and the MFT

A record for every file; journaling, smaller clusters, Unicode and hidden system files.

3

Inside an MFT record

Header, then attributes: 0x10 dates, 0x30 name and dates, 0x80 data.

Look in the file system's records, not just at the file.

UP NEXT

Slack Space and Deleted Files

Next, a hands-on lab: why deleted files survive, and how to recover one.

- **Slack space**
- **What deleting really does**
- **Recovering a deleted file**

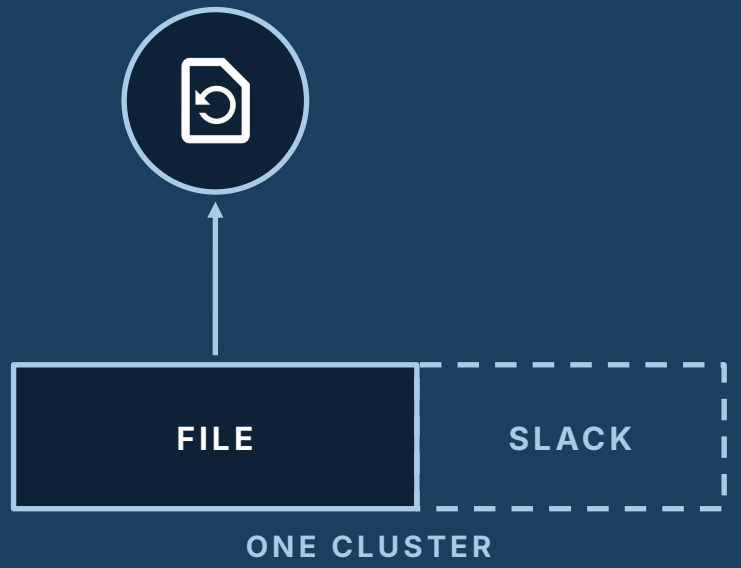
DFBOOK DIGITAL FORENSICS · LESSON 4.3

Slack Space and Deleted Files

Why deleted files survive, and how to recover one

IN THIS LESSON

1. Slack space
2. What deleting really does
3. Recovering a deleted file



LESSON 4.3 · OVERVIEW

Slack Space and Deleted Files

Deleting a file rarely removes it. The space is marked free, and the data waits until something new is written over it. In this lab you recover one.

BY THE END YOU WILL BE ABLE TO

1. **Explain drive slack, RAM slack and file slack**
2. **Explain what happens when a file is deleted**
3. **Recover a deleted file from an image**

INTRODUCTION · WHY THIS MATTERS

Why This Matters

What pressing Delete changes, and what it leaves behind

WHAT CHANGES

The records

- The file's entry is marked as deleted
- Its space is marked free for new data

WHAT STAYS

The data

- The contents remain in the same clusters
- Until a new file is written over them

A user who deletes invoices before an audit may leave them recoverable for weeks.

INTRODUCTION · ROADMAP

Your Route Through Lesson 4.3

Each part answers one question, in order

1

PART 1

Slack space

What is left over at the end of a file?

2

PART 2

What deleting really does

What changes on the disk when a file is deleted?

3

PART 3

Recovering a deleted file

How do you get it back from an image?

One case runs through the module: Teratai Supplies, where auditors have found payments for materials that were never delivered.

PART 1 OF 3

Slack Space

The unused space between a file's end and its cluster's end

01

IN THIS PART YOU WILL LEARN TO

- Explain drive, RAM and file slack
- Calculate slack for one file

KEY TERMS

Drive slack **RAM slack** **File slack**

PART 1 · STRUCTURE

Drive Slack

Files are given whole clusters, so the last cluster is rarely full



ONE CLUSTER, FROM START TO END

Drive slack	All the unused space in a cluster, after the end of a file.
RAM slack	Fills the rest of the file's last sector. Older Windows filled it from memory, so it could hold passwords; newer Windows writes zeros.
File slack	The remaining sectors of the cluster. They can still hold data from an earlier file.

PART 1 · WORKED EXAMPLE

Slack for One Small File

Illustrative: a 5,000-byte document on an older FAT16 drive with 32 KB clusters

THE FACTS

File size

`5,000 bytes`

Sector size

`512 bytes`

Cluster

`64 sectors = 32,768 bytes`

1. **Sectors the file touches**

   `5,000 / 512 = 9.8, so 10 sectors (5,120 bytes)`

2. **RAM slack**

   `5,120 - 5,000 = 120 bytes`

3. **File slack**

   `32,768 - 5,120 = 27,648 bytes`

ANSWER

Over 27,000 bytes of slack. NTFS uses smaller clusters, so it leaves far less.

PART 2 OF 3

What Deleting Really Does

The records change; the data usually stays

02

IN THIS PART YOU WILL LEARN TO

- Explain deletion on FAT and NTFS
- Judge whether a file is recoverable

KEY TERMS

Unallocated space **Recycle Bin**

PART 2 · SEQUENCE

Deleting a File on FAT

Two small changes to the records, none to the data

First letter replaced

The name's first character becomes hex E5, which marks the entry as deleted.

Chain cleared

The file's chain in the allocation table is set to 0.

Data untouched

The clusters become unallocated space, but keep their contents.

`INV1042.PDF` → `σNV1042.PDF`

How a tool may show the deleted entry: the first letter replaced by the E5 marker.

Unallocated space: space marked free for new data. It often still holds deleted files.

PART 2 · CONTRAST

Deleting a File on NTFS

Two routes: the Recycle Bin, or straight to free space

FROM FILE EXPLORER

To the Recycle Bin

- The file is renamed and moved to the Recycle Bin
- Its original name, path and deletion time are kept
- It can be restored, as Lesson 4.4 shows

COMMAND LINE OR EMPTIED BIN

Straight to free space

- Its clusters are marked free in $Bitmap
- Its MFT record is marked available
- Its run list is removed from the record

In both cases the contents usually remain until overwritten. NTFS reuses free space more efficiently than FAT.

PART 2 · WORKED EXAMPLE

Can These Invoices Be Recovered?

Illustrative: three deleted invoices, three different histories

What happened	Likely outcome
Deleted yesterday on a hard disk	Probably recoverable: little new data written since
Deleted, then a large video saved	Possibly partly overwritten: fragments may remain
Deleted on an SSD with TRIM on	May already be erased by the drive itself



ANSWER

Time, new writes and the type of drive decide what survives. Acquire early, and never write to the evidence.

PART 3 OF 3

Recovering a Deleted File

The lab: the Teratai invoice, in Autopsy

03

IN THIS PART YOU WILL LEARN TO

- Recover a deleted file from an image
- Record and verify what you recover

KEY TERMS

Recover **Extract** **Export**

PART 3 · CONCEPT

How Forensic Tools Recover Files

Three sources of deleted data, from easiest to hardest

1

Deleted entries

The record survives, so the name, dates and clusters can still be read.

2

Unallocated space

No record left; tools look for known file patterns in free clusters.

3

Slack space

Fragments of older files at the ends of clusters.

Defragmenting can leave old copies of a file in free space. Recovering from unallocated space, called carving, is covered in Lesson 5.2.

PART 3 · LAB

Recovering the Deleted Invoice

Case thread: the Teratai training image, opened in Autopsy

1. **Check the image**
   Confirm its hash matches the value recorded at acquisition.
2. **Open the case**
   Add the verified image to a new Autopsy case.
3. **Find deleted files**
   Under Views, open Deleted Files and look for the invoice.
4. **Look before you take**
   Check its name, path, dates and contents in the viewer.
5. **Extract it**
   Right-click and extract it to your case export folder.
6. **Hash and record**
   Hash the recovered file; note its path, dates and hash.

⚠ **Work only on the verified training image, and save recovered files to your export folder.**

PART 3 · METHOD

Recording a Recovered File

What your examiner notes must show for every recovered file

Where it was	The original path, and whether it came from a deleted entry or free space.
Its timestamps	Created, modified and accessed, as the tool reports them.
Its hash	So the recovered copy can be checked again later.
How it was recovered	The tool, its version and the steps you took.

LESSON 4.3 · SUMMARY

Lesson Summary

Three ideas that lead to one habit

1

Slack space

Unused space at the end of a cluster; RAM slack and file slack may hold old data.

2

What deleting does

FAT marks the name with E5; NTFS moves files to the Recycle Bin or frees them. Data stays.

3

Recovering a file

From deleted entries, free space or slack; extract, hash and record.

Deleted is not gone: look before you conclude.

UP NEXT

Windows Artefacts Investigators Use

Next, a second lab: the Recycle Bin, browser history and the Registry.

- **The Windows Registry**
- **Memory files and web history**
- **The Recycle Bin**

DFBOOK DIGITAL FORENSICS · LESSON 4.4

Windows Artefacts Investigators Use

Traces Windows keeps about what a user did

IN THIS LESSON

1. The Windows Registry
2. Memory files and web history
3. The Recycle Bin



LESSON 4.4 · OVERVIEW

Windows Artefacts

Windows records a great deal about how a computer is used: who logged on, which files were opened, which sites were visited, and what was deleted. These records are called artefacts.

BY THE END YOU WILL BE ABLE TO

1. **Say what the Registry may hold**
2. **Describe the pagefile, hibernation file and web history**
3. **Read a Recycle Bin record**

INTRODUCTION · WHY THIS MATTERS

Why This Matters

Files show what exists; artefacts show what a user did

Who was logged on?

User accounts and their settings

What did they open?

Recently used files and devices

Where did they browse?

Web history, downloads and forms

What did they delete?

Recycle Bin records, with times

Artefact: a trace an operating system or program leaves behind as a side effect of use.

In a fraud case, artefacts can link a person to the files and actions that matter.

INTRODUCTION · ROADMAP

Your Route Through Lesson 4.4

Each part answers one question, in order

1

PART 1

The Windows Registry

What does the Registry record about users?

2

PART 2

Memory files and web history

What do the pagefile, hibernation file and browser keep?

3

PART 3

The Recycle Bin

What does a deleted file's record reveal?

One case runs through the module: Teratai Supplies, where auditors have found payments for materials that were never delivered.

PART 1 OF 3

The Windows Registry

The database of settings and user activity

01

IN THIS PART YOU WILL LEARN TO

- Describe how the Registry is organised
- Say what evidence it may hold

KEY TERMS

Registry **Key** **Hive**

PART 1 · STRUCTURE

How the Registry Is Organised

A database arranged like folders and files

- **HKEY** — A top-level category
  - **Key** — Like a folder
    - **Subkey** — A key inside a key
      - **Value** — A name and its data, like a file

Hive: a branch of the Registry saved as a file on disk. Each user has a hive in a file named Ntuser.dat.

THE MAIN HKEYS

`HKEY_LOCAL_MACHINE`
Installed hardware and software

`HKEY_USERS`
Settings for each user account

`HKEY_CURRENT_USER`
The logged-on user's settings

PART 1 · CATEGORIES

Registry Files on Disk

Where to find the Registry in an image of a Windows drive

File and location	What it holds
Users\<account>\Ntuser.dat	Recently used files and the user's settings
...\config\SAM	User accounts and security settings
...\config\Software	Installed programs and their settings
...\config\System	System settings, including connected devices
...\config\Security	The computer's security settings

The config files sit in Windows\System32\config.

PART 1 · CONCEPT

What the Registry Can Show

Examples of evidence an examiner may find

- **The last user to log on**
- **Recently opened files**
- **USB devices connected**
- **Programs installed and used**
- **When a printer was used**
- **Networks and websites**

Examine the Registry from the image with a forensic tool, such as Autopsy or FTK Registry Viewer.

⚠ **On a live computer, never change a Registry setting: it can corrupt the system and alter the evidence.**

PART 2 OF 3

Memory Files and Web History

Traces of what was running, and where the user went

02

IN THIS PART YOU WILL LEARN TO

- Describe the hibernation and page files
- List what web history can show

KEY TERMS

hiberfil.sys **pagefile.sys**

PART 2 · CONTRAST

Memory Saved to Disk

Two hidden files that can hold what was in memory (RAM)

HIBERFIL.SYS

The hibernation file

- Written when a laptop lid closes or the PC hibernates
- Holds the contents of RAM at that moment
- Only the latest hibernation is kept

PAGEFILE.SYS

The page file

- Used when programs need more memory than the PC has
- Holds parts of memory moved out of RAM
- May sit on its own partition

Both are hidden files in the root of C:. Reading them needs memory-analysis tools, covered in a later module.

PART 2 · CATEGORIES

What Web History Can Show

Browsers collect a great deal about their users

- **Sites visited**
- **Downloads**
- **Bookmarks**
- **Cached pages**
- **Autocomplete text**
- **Saved form data**
- **Saved logins**
- **Add-ons**

Most forensic tools extract these items into a report. Your job is to explain, in plain words, how each one relates to the case.

For example: a visit to a supplier's site just before an invoice was approved.

PART 3 OF 3

The Recycle Bin

What Windows keeps about every deleted file

03

IN THIS PART YOU WILL LEARN TO

- Describe how the Recycle Bin stores files
- Interpret the fields of a $I record

KEY TERMS

$Recycle.Bin **$I file** **SID**

PART 3 · STRUCTURE

Inside the Recycle Bin

Each deleted file becomes a pair of files, in a folder for its user

MyTextFile.txt

Deleted in File Explorer

A folder per user

Named with the user's security identifier (SID)

Two new files

$I for its details, $R for its contents

```
$IXNIAC2.txt   details
```

```
$RXNIAC2.txt   contents
```

Not everything goes to the bin: Shift+Delete, or emptying the bin, removes these references.

PART 3 · STRUCTURE

The Fields of a $I Record

Four fields, at fixed positions (offsets) in the file

Offset	Field	What it shows
0x00	Windows version	02 for Windows 10 or 11
0x08	File size	The original size, in hexadecimal
0x10	Deleted date and time	**When the file was deleted**
0x1C	Path and file name	**Where the file was, and its original name**

A free command-line tool, RBCmd, reads $I files exported from an image and lists their fields in a spreadsheet file.

PART 3 · WORKED EXAMPLE

Reading One $I Record

Illustrative: a record found in a user's Recycle Bin folder

WHAT THE TOOL SHOWS

Folder (SID)
`S-1-5-21-...-1004`

Size
`48,213 bytes`

Deleted
`3 March, 18:42`

Original path
`C:\Users\clerk\Q1 notes.xlsx`

1. **Whose bin was it?**
   The account with that SID: match it to a user name.
2. **What was deleted?**
   Q1 notes.xlsx, from that user's own folder.
3. **When?**
   3 March at 18:42, by the computer's clock.

ANSWER

Who, what and when, from one small file. Check the clock and time zone before relying on the time.

PART 3 · LAB

Recycle Bin and Browser History

Case thread: the suspect account on the Teratai training image

1. **Find the account**
   Match the user's SID to their user name in Autopsy.
2. **Open their bin**
   Go to $Recycle.Bin and the folder for that SID.
3. **Read each $I file**
   Record the original path, size and deletion time.
4. **Open the $R file**
   Check the contents match the $I details.
5. **Review web history**
   Use Autopsy's results for that account's browser.
6. **Record and report**
   Note every artefact, its location and why it matters.

⚠ **Examine only the account in scope. Record other users' data only if it is relevant and permitted.**

LESSON 4.4 · SUMMARY

Lesson Summary

Three ideas that lead to one habit

1

The Windows Registry

Keys, values and hives; Ntuser.dat and config files show users, files and devices.

2

Memory files and web history

hiberfil.sys and pagefile.sys hold memory; browsers keep visits, downloads and more.

3

The Recycle Bin

A $I and $R pair per file, in a folder per user: path, size and deletion time.

Artefacts link a person to an action: record where each one came from.

UP NEXT

Lesson 4 Summary

Next, a short review brings Lessons 4.1 to 4.4 together before Lesson 5.



Drives and partitions



FAT and NTFS

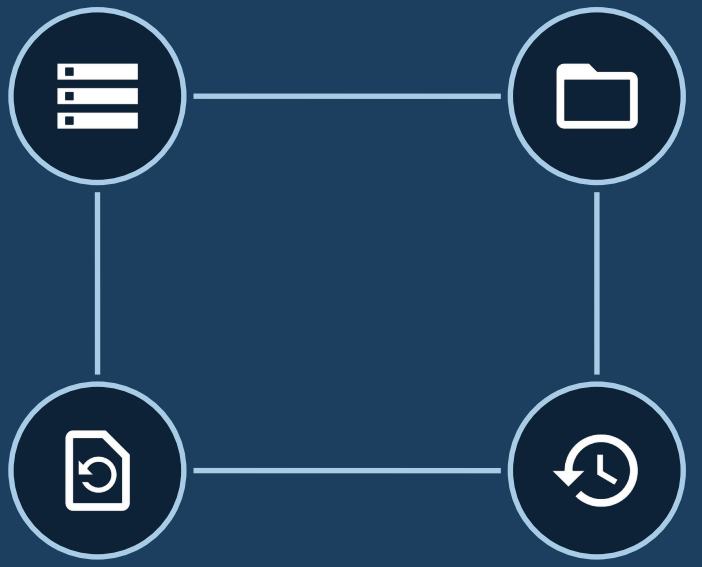


Deleted files and artefacts

DFBOOK DIGITAL FORENSICS · LESSON 4 · SUMMARY

Where Data Lives: Lesson Review

Disks, file systems, deleted files and artefacts, in one short review



IN THIS LESSON

1. Disks and file systems
2. Deleted files and slack
3. Windows artefacts

LESSON 4 · REVIEW

Lesson 4 at a Glance

One key point from each sub-lesson

4.1 Sectors to partitions	Sectors make clusters; partitions sit in MBR or GPT tables, and can be hidden.
4.2 FAT and NTFS	FAT chains clusters; NTFS keeps an MFT record, with two sets of timestamps.
4.3 Deleted files	Deleting changes records, not data; slack and free space can hold old files.
4.4 Artefacts	Registry, memory files, web history and the Recycle Bin show what a user did.

LESSON 4 · REVIEW

From Disk to User

How the four sub-lessons fit together



Deleting changes the records; the data usually stays.

UP NEXT

Examining Files and Finding Hidden Data

Lesson 5 turns stored data into findings, and a first timeline.

- **File signatures and metadata**
- **Searching and carving**
- **Filtering and timelines**

